Jinhyung Lee, Ph.D.

Postdoctoral Research Associate | Biomedical Engineering | University of Connecticut
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U.S. permanent resident; no visa sponsorship required

Research interests: Biomaterials and nanomedicine; RNA therapeutics; immunoengineering; gene regulation and genome editing.

Selected Research and Leadership Achievements

- **Principal Investigator, ORS/Bethel Fellowship Grant ($20,000; 2026–2027).** Lead studies of Atsttrin mRNA delivery in a human cartilage-on-a-chip model of osteoarthritis.
- **Seven peer-reviewed articles; four as first, co-first, or co-corresponding author.** First author, *PNAS* (2021); co-first author, *Journal of Controlled Release* (2025); co-corresponding author, *Materials Today Advances* (2026).
- **Technology translation:** Co-inventor on two granted patents and two published U.S. patent applications. U.S. Patent 12,214,043 B2 is licensed by UConn to Eascra Biotechnologies.
- **Research leadership:** Initiated an Atsttrin collaboration with Chuan-Ju Liu at Yale School of Medicine; coordinating investigator for a multi-institutional International Space Station spatial omics collaboration.

Proposed Independent Research Program

Define how nanoparticle geometry and receptor targeting govern functional RNA delivery in immune-mediated disease. Initial studies will focus on macrophage-directed RNA therapeutics for rheumatoid arthritis, with staged extensions to CNS delivery and cell-restricted gene regulation.

Education

Ph.D., Biomedical Engineering — University of Connecticut 2024
M.S., Bioengineering — University of California, San Diego 2017
B.S., Convergence Biotechnology — Konkuk University 2015

Research Appointments

Postdoctoral Research Associate, University of Connecticut 09/2024–present
Lead folate-directed Janus base nanoparticle studies for RNA delivery to inflamed synovium and cartilage.
Collaborate with Dr. Eun Ji Chung at the University of Southern California (USC) on gene editing in the kidney.
Research Assistant, University of Connecticut 08/2019–05/2024
Researcher, Brain Science Institute, KIST 09/2017–12/2018
Graduate Research Assistant, UC San Diego 01/2016–06/2017

Teaching and Mentoring

Teaching: Teaching Assistant, Foundations of Biomechanics (BENG 226; graduate level), UC San Diego, 2017; led discussions and developed supplementary materials.
Mentoring: Supervised seven graduate and undergraduate researchers since 2019; trained mentees in nanoparticle characterization and RNA-delivery methods. Trainees coauthored four conference presentations.
Teaching interests: Biomaterials; Drug Delivery Systems; Nanomedicine; Tissue Engineering; Introduction to Biomedical Engineering; Biomechanics.

PEER-REVIEWED PUBLICATIONS

7 articles | **Lee J** denotes Jinhyung Lee | † Equal contribution | * Corresponding author

1. †Zhai J, †Surian A, Cote T, Zhang W, Dang BT, Wang Y, Chen Q, Hoshino K, *†**Lee J**, *Chen Y. "Enhanced endosomal escape for nanoparticle-enabled co-delivery of doxorubicin and siRNA to overcome multidrug resistance." *Materials Today Advances*. 2026;30:100845. doi:10.1016/j.mtadv.2026.100845
2. †**Lee J**, †Zhang W, Nguyen D, Zhou L, Amengual J, Zhai J, Cote T, Landolina M, Ahmadi E, Sands I, Mishra N, Yu H, Nieh MP, *Wang K, *Li Y, *Chen Y. "Computation-aided design of rod-shaped nanoparticles for tumoral targeting." *Journal of Controlled Release*. 2025;387:114169. doi:10.1016/j.jconrel.2025.114169
3. Zhou L, Zhang W, **Lee J**, Kuhn L, *Chen Y. "Controlled self-assembly of DNA-mimicking nanotubes to form a layer-by-layer scaffold for homeostatic tissue constructs." *ACS Applied Materials & Interfaces*. 2021;13(43):51321–51332. doi:10.1021/acsami.1c13345
4. **Lee J**, Sands I, Zhang W, Zhou L, *Chen Y. "DNA-inspired nanomaterials for enhanced endosomal escape." *Proceedings of the National Academy of Sciences USA*. 2021;118(19):e2104511118. doi:10.1073/pnas.2104511118
5. †Yau A, †**Lee J**, *Chen Y. "Nanomaterials for protein delivery in anticancer applications." *Pharmaceutics*. 2021;13(2):155. doi:10.3390/pharmaceutics13020155
6. Sun M, **Lee J**, Chen Y, Hoshino K. "Studies of nanoparticle delivery with in vitro bio-engineered microtissues." *Bioactive Materials*. 2020;5(4):924–937. doi:10.1016/j.bioactmat.2020.06.016
7. Sands I, **Lee J**, Zhang W, *Chen Y. "RNA delivery via DNA-inspired Janus base nanotubes for extracellular matrix penetration." *MRS Advances*. 2020;5(16):815–823. doi:10.1557/adv.2020.47

PATENTS AND TECHNOLOGY TRANSFER

1. Chen Y, **Lee J**, Zhang W. "Therapeutic nanomaterials." U.S. Patent 12,214,043 B2; granted February 4, 2025. Prior application US20220133893A1 published May 5, 2022. Licensed to Eascra Biotechnologies.
2. Chen Y, Zhang W, **Lee J**. "Nanomaterial and methods of use thereof." U.S. Patent Application Publication US20250367300A1; published December 4, 2025.
3. Chen Y, Zhang W, **Lee J**. "Nanomaterial delivery vehicle and method of use thereof." U.S. Patent Application Publication US20250302762A1; published October 2, 2025.
4. Seong J, **Lee J**, Kim H, Choo H. "Biosensor based on fluorescence resonance energy transfer for detecting of antipsychotic medicine and detection method for using same." Korean Patent 10-2176538 B1; registered November 3, 2020; published November 9, 2020.

SELECTED PRESENTATIONS AND PROFESSIONAL SERVICE

1. **Lee J**, Cote T, Dang BT, Zhai J, Liu CJ, Chen Y. "Long-acting DNA-inspired Janus base nanoparticle depot for intra-articular mRNA therapy in osteoarthritis." **Accepted podium presentation**, ORS Annual Meeting, San Francisco, CA, February 21, 2027 (upcoming).
2. **Lee J**, Cote T, Zhai J, Dang BT, Chung EJ, Landolina M, Chen Y. "Rod-shaped DNA-inspired Janus base nanoparticles enable in vivo kidney CRISPR editing." **Accepted oral presentation**, BMES Annual Meeting, Orlando, FL, October 2026.
3. **Lee J**, Cote T, Davidson N, Zhai J, Zhang W, Morgan K, Liu C, Laurencin C, Chen Y. "Janus base nanopieces delivered mRNA therapeutics for osteoarthritis treatment." **Podium presentation**, Orthopaedic Research Society Annual Meeting, 2024.

Invited talks: Chonnam National University Medical School (2026); Biomaterials and Biofabrication Conference (2025); Kyung Hee University College of Medicine (2024); Next-Generation Nanomedicine (2023).

Journal reviewer: *ASSAY and Drug Development Technologies* (2026).